

Database Systems

CSC 4710/6710: Spring 2026
Department of Computer Science
Georgia State University
Jan 12, 2026

Class Time & Location

Monday Wednesday 10:00 AM - 11:45 AM.
Classroom South | Room 103

Instructor Information

- Instructor: Farhan Tanvir
- Email: ftanvir@gsu.edu
- Office hours & Location: Tuesday Thursday 12:00 PM – 2:00 PM, 1 Park Place 717
<https://gsumeetings.webex.com/meet/ftanvir>

TA Information:

Truong Le
Email: sle22@student.gsu.edu
Office hours: Wednesday 12pm-2pm (it can be online or face-to-face at Cubicle 639/640)

Course Outcomes

Upon successful completion of this course of study, the student will be familiar with:

- The basic principles of designing and implementing an application using relational database systems;
- How a relational database application works for designing (the schema for), mapping and querying structured data;
- Learning to design ER diagram;
- The database language SQL and relational SQL;
- Development of a web-based database system development

Course Description

As the first introductory course for databases, it studies the fundamentals of relational database management systems (RDBMS). Materials include, but are not limited to, the ER (Entity-Relationship) model and its mapping to the relational data model, the algebraic language in the relational data model and its relationship to the commercial relational query

language: SQL, database design that conforms to different normal forms, database applications, database indexing, transaction management, recovery and other advanced topics. Furthermore, students will participate in a semester-long project and build a web-based database system. In summary, this course is about the principles of designing and developing applications using relational database systems.

Textbook

Required: Zybooks

Code: GSUCSC47106710DSCI4710TanvirSpring2026

In this course, we will use zybooks, which is an online textbook and assignment system. Most of the assignments will be covered through zybooks, so it is required to subscribe on zybooks.

[recommended] Database System Concepts, Abraham Silberschatz, Henry F. Korth, S. Sudarshan, 7th Edition, McGraw-Hill / ISBN 978-0078022159

Database Illuminated, Catherine M. Ricardo and Susan D. Urban, 4th Edition, Jones & Bartlett Learning / ISBN 9781284231588

Prerequisites

CSC 2720 with a grade of C or higher (for CSC 4710) and CSC 3410 (for CSC 6710). Intermediate programming skills and an understanding of basic mathematical skills are expected.

Students should come with good programming skills.

If you are not sure whether you have the right background, please contact the instructor.

Note: We will not cover programming-specific issues in this course.

Format and activities

The course is lecture-based with three mid-terms examinations, assignments and projects. There are individual assignments and a group-level programming project.

- Lectures and Class Participation: We strongly encourage (and appreciate!) students to attend classes, because effective lectures rely on students' participation to raise questions and contribute in discussions. **10% of final grade will come from attendance. Attendance won't be given if student is 20 minutes late.** We will strive to maintain interactive class discussions if possible. We will provide lecture notes before class, which will be posted on the on ICollege.
- Questions: We encourage students to discuss their questions and problems first with their group peers and classmates. This way, you can get immediate help and also learn to communicate "professionally" with your peers. In any case for a more thorough discussion,

attend the office hours of TA and the instructor. Any announcement will be posted on the ICollege. Make sure to check it frequently enough to stay informed.

- Assignment: There will be 2 written and few zybook assignments spaced out over the course of the semester. All the assignments should be done individually by the students.
- Exam: There will be 2 midterm exams. No Final exam. All exams should be done individually by the students.
- Projects: There will be a semester-long project, which involves significant database application programming. The project will be structured with several milestones due in the course of the semester, leading to a demo and write-up near the end of the semester.

Grading Policy

Requests for regrading should be within a week of grades being posted on ICollege. The final course grade will be computed as follows:

Midterms	40%
Project	35%
Written Assignments	10%
Zybook Assignments	5%
Attendance	10%

A+	A	A-	B+	B	B-	C+	C	C-	D	F
97-100	93-97	90-93	87-90	83-87	80-83	77-80	73-77	70-73	60-70	0-60

Late Assignment Policy

- There will be 15% penalty for 1-week late submission. There will be 30% penalty for 2-week late submission. **More than 2 week late assignments will not ordinarily be accepted.** If, for some compelling reason, you cannot hand in an assignment on time, please contact the TA or instructor as far in advance as possible. Written assignments or project deliverance are due at the beginning of a class, you should hand them in at the beginning of the class;
- No make-up exams (except under extremely unusual circumstances).

Communication Policy

E-mail is an appropriate and preferred method for communication. I will communicate with you by e-mail with the assumption that you will receive, read and, if necessary, act in a timely manner based upon these e-mails. I will also often communicate with you by posting announcements on our class iCollege page.

If I need to contact you outside of class, I will use your GSU email address, and expect that you will read and respond to this communication in a timely manner.

Please recognize standard email etiquette. Initial emails to me should contain (minimally) a subject containing "CS 4710/6710", greeting, clear question/inquiry, and closing. Brief and to-the-point emails are appreciated.

I will attempt to respond to students within 1 business day. If you have not received a reply from me within 2-3 business days, please resend the email. Please be respectful to your instructor and teaching team as you communicate.

Academic Dishonesty

The Computer Science departmental policy for academic dishonesty and misconduct applies to this class. In addition, a student attempting to gain an unfair advantage by keeping an examination paper longer than the time permitted is guilty of academic misconduct.

Assignments/projects/exams are to be done individually unless specified otherwise. It is a violation of the Academic Honor Code to take credit for the work done by other people. It is also a violation to assist another person in violating the Code. Examples of cheating behaviors include:

- Copy programs for programming assignments.
- Turning in group work for assignments where you are expected to work as an individual.
- Use and submit existing programs/reports on the world wide web as written assignments.
- Submit programs/reports/assignments done by a third party, including hired and contracted.
- Plagiarize sentences/paragraphs from others without giving the appropriate references.

Accommodation for Disabilities

The Computer Science departmental policy for students with disabilities applies to this class. Anyone who has a need for examinations by special arrangements should see the instructor as the earliest possible opportunity during scheduled office hours.

Syllabus Change Policy This syllabus is a tentative guide for the course and is subject to change. You'll be informed in class if there's a change in the syllabus.